SCHOOL OF TECHNOLOGY BACHELOR OF SCIENCE IN SOFTWARE DEVELOPMENT

UNIT: EMBBEDED SYSTEMS 3205

AUTOMATIC MEDICINE REMINDER PROJECT

GROUP MEMEBERS

Otieno Javis 22/03328
Achan Ayiik 21/01327
Mitemah Daphne Grace 21/08363
Mark Kinoti 19/06839
Samwel Gaya Odhiambo 22/07809
Jeylani Ibrahim 22/07343
Akram Abdirahman 22/08232

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ABBREVIATIONS

RTC – Real-Time Clock

LCD – Liquid Crystal Display

LED – Light Emitting Diode

IoT – Internet of Things

EEPROM – Electrically Erasable Programmable Read-Only Memory

KES – Kenyan Shillings

UNO – Universal Network Object (Arduino model)

BoM – Bill of Materials

Project Overview
Automatic Medicine Reminder is intended to assist individuals, especially the elderly or persons with chronic illnesses, in being able to manage their medication schedule effectively. The device uses an Arduino microcontroller to remind users of taking the proper medicines at the appropriate times. It avoids skipped doses, overdosing, or taking the inappropriate medicines, which lead to critical health consequences. It has audible and visual cues and can be programmed for use with greater than one medication schedule. The classes of users include caregivers, healthcare workers, and patients, and the specific issue addressed is the inconvenience of adherence to intricate medication regimens with no aid.

Project Title

Automatic Medicine Reminder Using Arduino

Objective:

This project aims to develop a smart medicine reminder system that assists patients, especially the elderly, in adhering to their prescribed medication schedules. The system utilizes an Arduino microcontroller to trigger alerts using sound, visual indicators, and an LCD display when medication is due. By ensuring medication adherence, the system reduces health risks associated with missed doses.

Potential Applications:

- Home healthcare automation (smart medication scheduling)
- Patient assistance systems in hospitals and elderly care homes
- Assistive technology for people with cognitive impairments

Project Requirements

1. **Concept and Planning**
   - **Problem Statement:**
   Medication non-compliance is a significant issue among elderly patients, leading to deteriorating health conditions, hospitalization, or even life-threatening consequences. Traditional reminder methods, such as written schedules or mobile alarms, may not be effective for elderly individuals with cognitive impairments or disabilities. Our solution addresses this problem by providing an automated, user-friendly reminder system integrated with both auditory and visual alerts to ensure timely medication intake.

   - **Objectives and Scope:**
- Automate medication reminders using an Arduino-based system to ensure timely intake of prescribed medicines.
- Implement an alert system with visual (LED) and auditory (buzzer) notifications to remind users to take their medication.

- Enhance medication adherence by providing a reliable and customizable reminder system for users with complex medication schedules.

- Provide a user-friendly interface for setting and adjusting medication schedules, ensuring ease of use for individuals of all ages.

- Improve health outcomes by reducing the risk of missed or incorrect doses, particularly for elderly patients or those with chronic conditions.

- Design a portable and low-cost solution that is accessible and convenient for home or travel use.

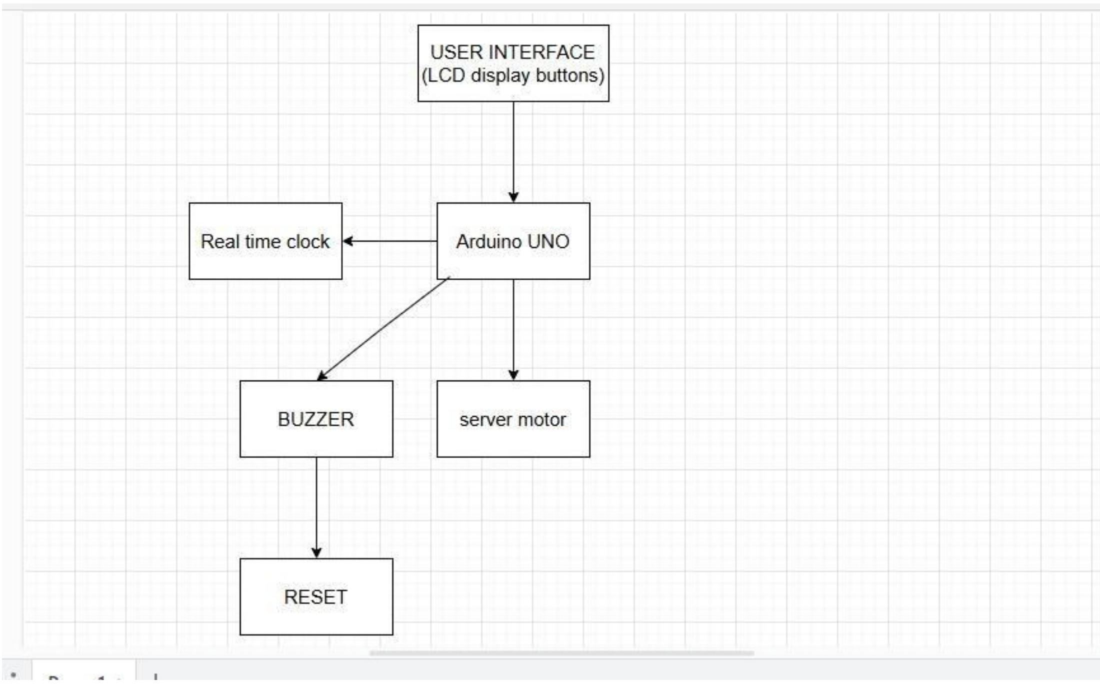
o **System Architecture and Block Diagram:**

The **Automatic Medicine Reminder Using Arduino** is designed to assist patients in taking medications on time. It integrates various components to provide audio-visual reminders and automated dispensing mechanisms.

Key Components:

1. **Arduino (Microcontroller)** – The core of the system that processes data and controls components.
2. **Real-Time Clock (RTC Module, e.g., DS3231)** – Maintains accurate time for scheduling reminders.
3. **LCD Display (16x2 or OLED)** – Shows the medicine name and dosage instructions.
4. **Buzzer/Speaker** – Provides an audible alert when it’s time to take medicine.
5. **LED Indicators** – Blinks to attract attention to the reminder.
6. **Push Button (or Touch Sensor)** – Used to confirm medicine intake and reset alarms.
7. **Servo Motor** – Controls the medicine dispensing mechanism (optional).
8. **Power Supply (Battery or Adapter)** – Provides power to the system.
9. **Mobile Notification (Optional: Bluetooth Module/IoT Integration)** – Sends reminders via a smartphone app.

Block Diagram



Justification for Components:

Arduino (Microcontroller)

Rationale: It is the system brain, processing data, managing inputs/outputs, and commanding all the parts.

Why Arduino? Easy to code, affordable, and supports many sensors and modules.

Real-Time Clock (RTC Module, i.e., DS3231)

Rationale: Keeps current time and triggers reminders at the right times.

Why DS3231? Accurate, minimal power consumption, and has battery backup to keep time even when there is a power outage.

LCD Display (16x2 or OLED)

Reason: Provides a visual reminder by displaying the name of medicine, dose, and time.

Why LCD? Low power usage and simple integration with Arduino.

Why OLED (Optional)? Enhanced contrast, readability, and provides support for graphical data.

Buzzer/Speaker
Reason: Provides an audio reminder to remind the user to take medicine at the proper time.
Why a Buzzer? Simple, cheap, and can generate a high-intensity signal.

Why a Speaker (Optional)? Can read out voice reminders instead of beeping.

LED Indicators

Justification: Visual cue to attract the user, useful for hearing-impaired users as well.

Why LEDs? Low power consumption, long life, and easy integration.

Push Button (or Touch Sensor)

Justification: For acknowledgement by user of taking medicine and resetting the alarm.

Why Push Button? Simple and sturdy for user input.

Why Touch Sensor (Optional)? Neater look and feel and with no mechanical wear.

Servo Motor (Optional)

Rationale: Can be used to automatically dispense medicine at the schedule time.

Why Servo? Provides precise control of small mechanical movements, ideal for pill dispensing.

Power Supply (Battery or Adapter)

Rationale: Ensures uninterrupted system operation.

Why Battery? Provides portability, easy for elderly users.

Why Adapter? Provides stable and uninterrupted supply, reducing maintenance.

Mobile Notification (Optional: Bluetooth Module/IoT Integration)

Rationale: Reminds by a smartphone app to notify caregivers or patients remotely.

Why Bluetooth?

Enables communication at short distances with a mobile app.

Why IoT/Wi-Fi (Optional)?

Enables remote monitoring, which is very handy for caregivers attending to several patients.

Timeline and Milestones:

Week 1-2: Research and design of system architecture.
Week 3-4: Assembly of hardware components and coding initiation.
Week 5-6: Implementation of software logic and unit testing.
Week 7: Debugging and optimization of the system. Week 8: Final testing and demonstration.

1. **Hardware and Software Components**

Hardware Requirements

- **Arduino Uno** – Chosen for its ease of use, availability, and sufficient I/O pins.
- **RTC DS3231 Module** – Provides precise timekeeping for medication scheduling.
- **16x2 LCD Display** – Displays medication alerts and settings.
- **Buzzer & LED** – Alerts the patient when medication is due.
- **Push Buttons** – Allows users to set and modify reminder times.
- **10K Potentiometer** – Adjusts LCD contrast.
- **Resistors (10K, 1K)** – Ensures circuit stability.
- **Jumper Wires & Breadboard** – Used for connectivity and trial wiring.

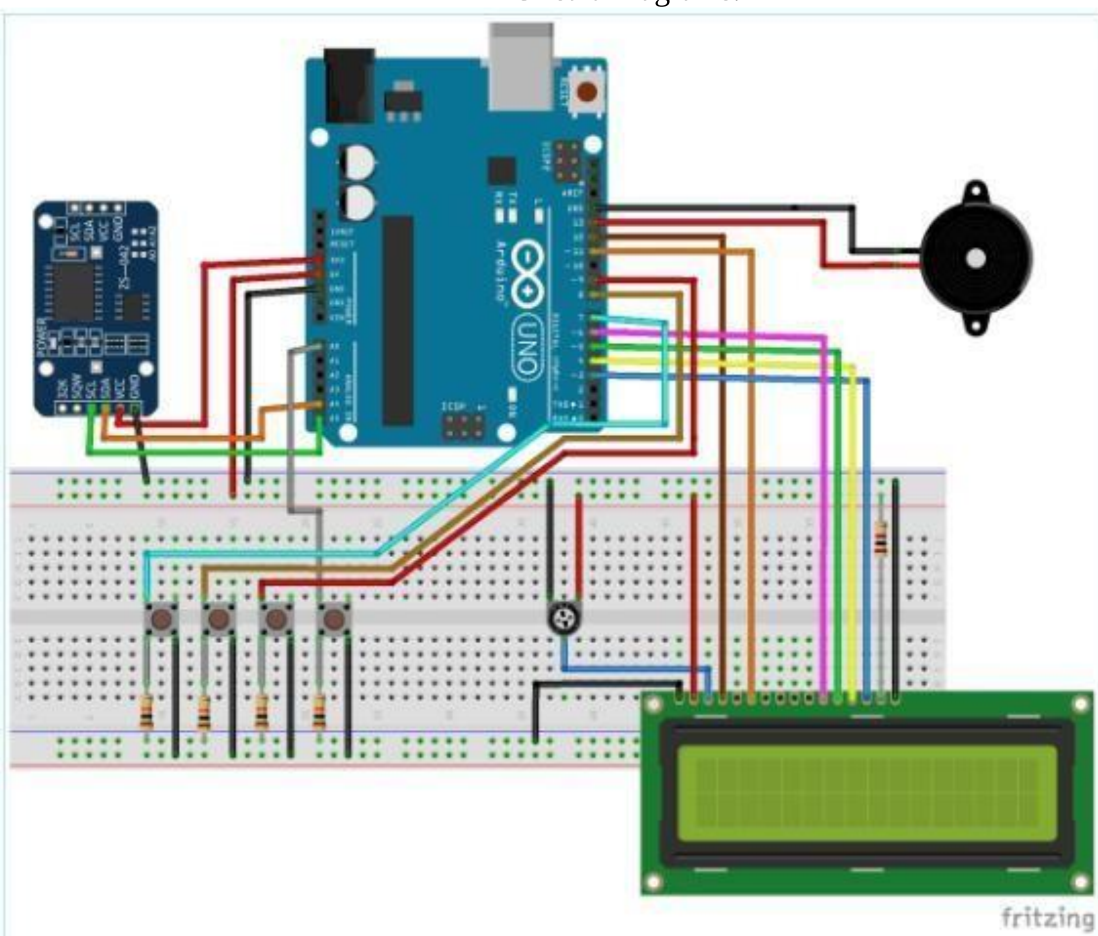
Hardware Budget:

Component	Description	Ksh
Arduino Uno	Microcontroller board with sufficient I/O pins.	1,500
RTC DS3231 Module	Real-Time Clock module for precise timekeeping.	300
16x2 LCD Display	Screen to display medication alerts and settings.	500
Buzzer & LED	Audio and visual indicators for medication alerts.	100
Push Buttons	User input for setting and modifying reminder times.	30
10K Potentiometer	Adjusts the contrast of the LCD display.	50
Resistors (10K, 1K)	Ensures circuit stability and proper current flow	15

Jumper Wires & Breadboard	Essential for prototyping and connecting components without soldering.	200

TOTAL: 2695 Ksh

Circuit Diagrams:



Software Requirements

 Development Environment – Arduino IDE
Programming Language – C **Libraries Used:**

o Wire.h (RTC communication)
o LiquidCrystal.h (LCD interaction)
o TimeLib.h (Time comparison functions)
Firmware Organization:
o Functions for setting reminders, checking time, generating alerts, and displaying messages.

o Modular coding approach with comments for readability and maintainability.

3. **Implementation and Testing**

Implementation Plan

1. **Phase 1 - Research & Design (Weeks 1-2):** Identify system components and develop initial architecture.
2. **Phase 2 - Hardware Assembly & Code Development (Weeks 3-4):** Assemble hardware and develop core firmware.
3. **Phase 3 - Feature Integration & Testing (Weeks 5-6):** Implement reminder setting and debugging.
4. **Phase 4 - Optimization & Final Testing (Weeks 7-8):** Conduct full testing and prepare for demonstration.

Testing Strategy

Unit Testing

- Verify RTC timekeeping accuracy. Test LCD display output.
- Ensure buzzer and LED activation.

System Testing

- Check interactions between hardware and software components.
- Validate correct alarm triggering.

Validation & Troubleshooting

- Confirm reminder accuracy.
- Debug potential failures.
- Implement error-handling routines. **Safety and Reliability**:

Proper wiring techniques to prevent short circuits.
Strong error handling for time sync failures.
Secure power management for long-term reliability.

4. **Demonstration and Documentation**

Live Demonstration

- Configure medication reminders and present real-time alerts.
- Explain system functionality and user interaction.

Project Report

- **Abstract** – Summary of the project.
- **Introduction & Background** – Importance of medication adherence.
- **Design & MethodoloConclusion & Future Work** – Recommendations for improvements.
- **gy** – Implementation details.
- **Results & Discussion** – System performance and challenges.
- **References** – Cited external sources.

Presentation

- PowerPoint slides showcasing hardware setup, flowcharts, and test results.
- Videos demonstrating system functionality.

Evaluation Criteria

1. Project Planning & Design
2. Hardware & Circuit Design
3. Software & Code Quality
4. Implementation & Testing
5. Demonstration & Documentation

Submission and Milestone Timeline

1. **Proposal Submission (Week 2)** – Concept submission with block diagram.
2. **Design Documentation (Week 4)** – Circuit diagrams and software structure.
3. **Prototype Demonstration (Week 6)** – Initial prototype and test results.
4. **Final Submission (Week 8)** – Full project report, final code, and live demo.

Additional Considerations

- **Innovation & Creativity** – Possible IoT integration.

- **Collaboration & Communication** – Defined team roles.
- **Backup & Recovery** – Regular GitHub commits.
- **Ethics & Safety** – Electrical safety compliance. **References**:

RTC DS3231 Datasheet
Arduino Official Documentation (https://www.arduino.cc/)
Sample projects from Instructables, Hackster.io, and IEEE Xplore

1.EXECUTIVE SUMMARY

Project Summary

Automatic Medicine Reminder is a project that intends to create a reminder for patients, especially those with chronic illnesses or the elderly, to take their medication. It aims to improve medication intake for these people for better health and improved recovery rate in patients. It also aims to reduce the mental load of remembering when to take your medication.

Problem It Solves.

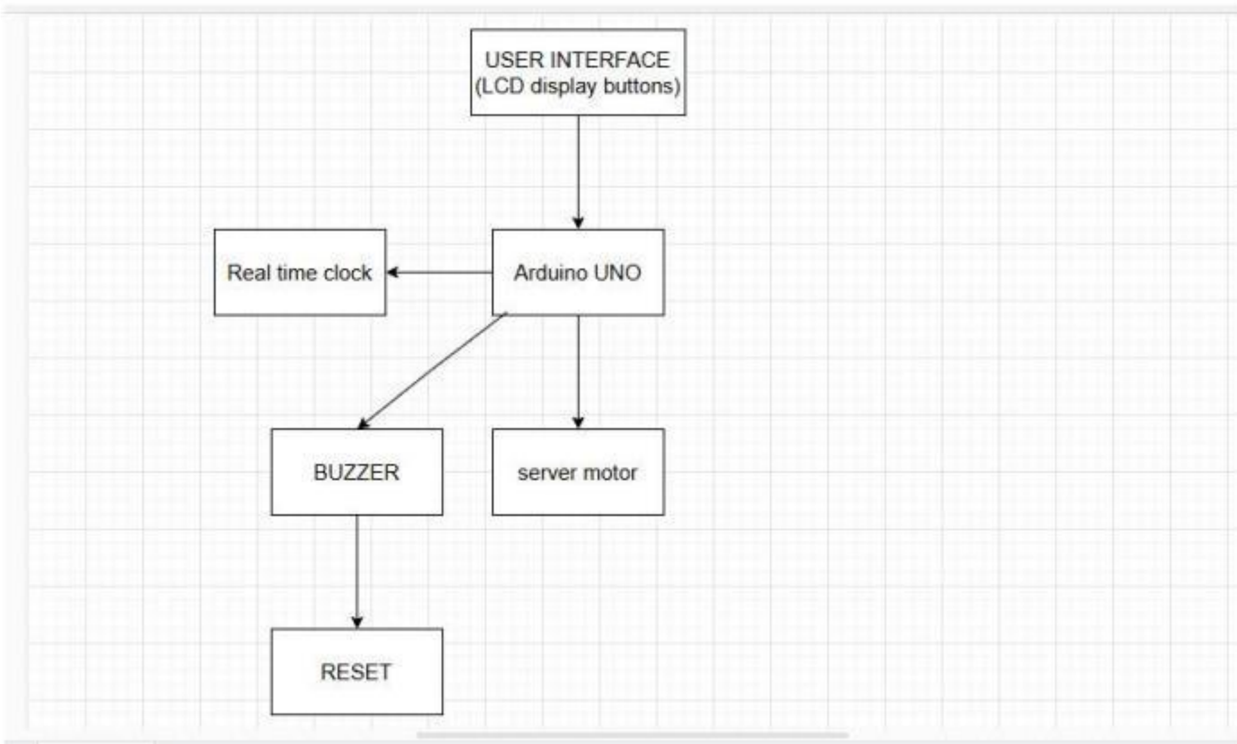
The project aims to solve the problem of underdosing, overdosing, or skipped doses, which may have dire consequences to the patient. The impact in turn is improved health and/or improved recovery rate for these individuals.

Key Features and Functionality

The following are the key features of the project and what they are supposed to do:

a) **Arduino (Microcontroller)** – The core of the system that processes data and controls b) components.
c) **Real-Time Clock (RTC Module, e.g., DS3231)** – Maintains accurate time for scheduling
d) reminders.
e) **LCD Display (16x2 or OLED)** – Shows the medicine name and dosage instructions.
f) **Buzzer/Speaker** – Provides an audible alert when it's time to take medicine.
g) **LED Indicators** – Blinks to attract attention to the reminder.
h) **Push Button (or Touch Sensor)** – Used to confirm medicine intake and reset alarms.
i) **Servo Motor** – Controls the medicine dispensing mechanism (optional).
j) **Power Supply (Battery or Adapter)** – Provides power to the system.
k) **Mobile Notification (Optional: Bluetooth Module/IoT Integration)** – Sends reminders via a smartphone.

2.SYSTEM OVERVIEW
Block Diagram



Component Integration

i. Sensor Data Collection: the touch sensor is touched to set or reset the time for the medication.

ii. Microcontroller: the Arduino reads the set time and processes it. When the set time has come the microcontroller makes a decision based on the codes.

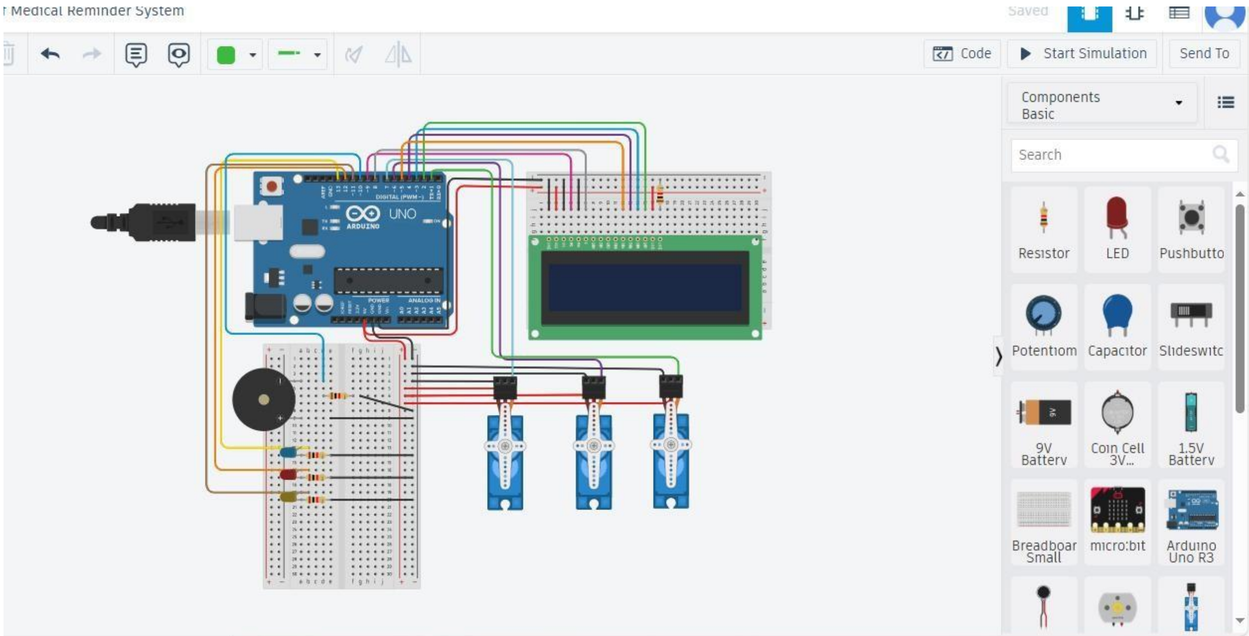
iii. Actuator: after the microcontroller has sensed that it is time for medication, it triggers the buzzer to go off and the LED indicators lights up.

iv. Communication Module: The LCD display then indicates the message on the screen for the user to see.

User interaction is handled by the push button. It is used to set the time for when the reminder is to go off. Also, the LED indicators, the buzzer, and the LCD Display help in output. They attract the user's attention to the reminder.

3. Hardware Implementation

Finalized circuit diagram for Arduino medicine reminder


Medical Reminder System
Saved
Code
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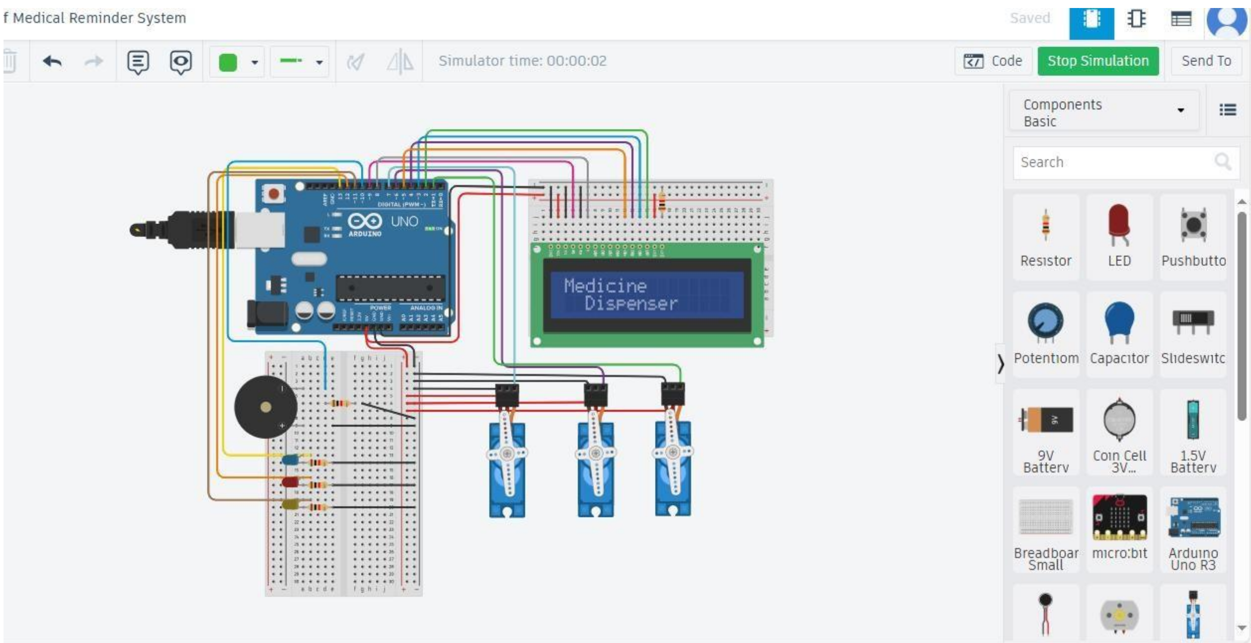
f Medical Reminder System
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Code
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Medicine
Dispenser
Components
Basic
Search
Resistor
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Capacitor
Slideswitc
9V
Battery
Coin Cell
3V...
1.5V
Battery
Breadboar
Small
micro:bit
Arduino
Uno R3

Component List

Download CSV

Name	Quantity	Component
SERVO1 SERVO2 SERVO3	3	Positional Micro Servo
U1	1	Arduino Uno R3
D1	1	Yellow LED
D3	1	Blue LED
D4	1	Red LED
U2	1	LCD 16 x 2
PIEZO2	1	Piezo
R1 R2 R3 R4 R5	5	1 kΩ Resistor

Bill of Materials (BoM):

Item No.	Part Number	Component Name	Quantity	Description	Supplier/Manufacturer	Price (KES)
1	SERV01, SERV02, SERV03	Positional Micro Servo	3	Small servo motors for movement	Komponent Electronics	300 each
2	U1	Arduino Uno R3	1	Microcontroller board	Komponent Electronics	1,500
3	D1	Yellow LED	1	Indicator LED (Yellow)	ASK Electronics	10
4	D3	Blue LED	1	Indicator LED (Blue)	ASK Electronics	10
5	D4	Red LED	1	Indicator LED (Red)	ASK Electronics	10
6	U2	LCD 16x2 Display	1	16x2 Character LCD Display	ASK Electronics	550
7	PIEZO2	Piezo Buzzer	1	Sound alert buzzer	ASK Electronics	50

8	R1, R2, R3, R4, R5	1kΩ Resistor	5	Resistors for circuit design	ASK Electronics	5 each

4. Software Implementation
Firmware Code

The Automatic Medicine Reminder system's source code has been thoroughly examined and documented. It can interface with parts like the servo motor (optional), LCD display, buzzer, LED indicators, RTC module, Arduino microcontroller, and push buttons. The code guarantees smooth hardware and software integration to deliver timely medication reminders.

Structure of Code

To improve readability and maintainability, the code is modularized:

Main Program: Controls the flow of control and initializes components.

Modules:

To maintain precise time, the RTC Module manages real-time clock functionality.

Display Module: Controls the LCD screen to display information about medications.

The alert module is in charge of the LED indicators and buzzer.

Push buttons are used by the input module to process user input.

External Libraries: Common Arduino libraries such as Wire. H for I2C communication with the RTC module and LiquidCrystal.h for managing the LCD display are used.

Control Algorithms

The control algorithm is based on time-matching logic:

The RTC module continuously monitors the current time.

Pre-defined medication times are compared with the current time.

When a match occurs, the system triggers an alert using the buzzer and LED indicators while displaying relevant information on the LCD screen.

The user confirms medication intake by pressing a button, resetting the alarm.

Error Handling

The system incorporates several error-handling mechanisms:

Component Failure Detection: Alerts are triggered if critical components (e.g., RTC or display) fail to initialize.

Time Synchronization Errors: The system reinitializes the RTC module if discrepancies in time are detected.

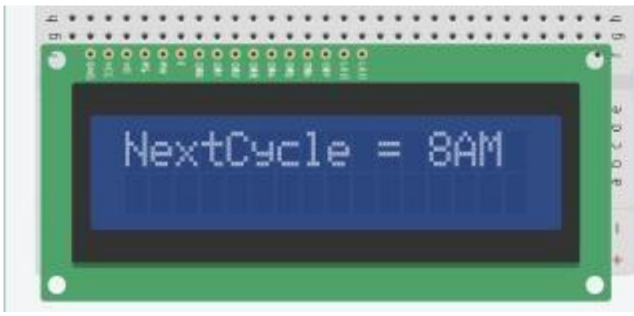
User Errors: If no input is received after an alert, a secondary reminder is issued.

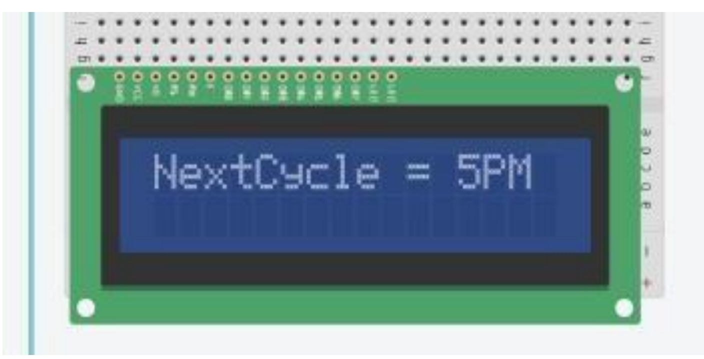
5. Testing and Validation

Functional Testing

Test Case	Expected Outcome	Actual Result	Status
RTC initialization	Accurate time displayed	Accurate	Pass
Medication alert at 8 AM	Buzzer sounds, LED blinks, info on LCD	As expected	Pass
User confirmation via button	Alert stops	As expected	Pass
Missed confirmation	Secondary alert issued	As expected	Pass

6. User Interface & Interaction





User Flow

a) **System Initialization**: The Arduino UNO powers on, initializes the LCD, LEDs, and buzzer, and displays a welcome message.

b) **Reminder Trigger**: At the scheduled medication time, the buzzer sounds, the red LED flashes, and the LCD displays the medication name and dosage.

c) **User Interaction**:

   a. If the user takes the medication, they press the **Acknowledge Button**, which stops the buzzer, turns off the red LED, and logs the event.

   b. If the user needs more time, they press the **Snooze Button**, which delays the reminder for 5 minutes.

d) **Error Handling**: If the user misses a medication, the red LED remains on, and the LCD displays a missed medication alert.

e) **System Logging**: All user interactions and missed reminders are logged for future reference.

Feedback Mechanisms

i. **Visual Feedback**:
   a. LCD displays clear messages for reminders, acknowledgments, and errors.
   b. LEDs provide quick status updates (green for normal operation, red for errors).

ii. **Audible Feedback**: The buzzer ensures the user is alerted even if they are not looking at the device.

iii. **Logs**: A simple log is maintained on the Arduino's EEPROM, which can be accessed via a serial monitor for debugging or review.

7. Final Demonstration Plan

Step-by-Step Demonstration of the Near-Complete System

1. **Introduction**
   o Briefly explain the purpose of the project and its real-world application.
   o Showcase the assembled prototype.
2. **System Power-On & Initialization** o Demonstrate turning on the system and initializing the Arduino.
   o Show the LCD display initializing and displaying system status.
3. **Setting Up Medication Reminders** o Configure a sample medication schedule using push buttons.
   o Confirm the set reminder appears on the LCD screen.
4. **Reminder Activation** o Wait for the scheduled medication time.
   o Demonstrate the buzzer sound, LED flashing, and LCD notification.
5. **User Acknowledgment & Alarm Reset** o Press the confirmation button to acknowledge medication intake.
   o Show that the alert stops and resets for the next scheduled time.
6. **(Optional) IoT/Bluetooth Integration** o If implemented, demonstrate a notification being sent to a caregiver's phone.
7. **System Shutdown & Summary**
   o Power off the system and summarize its functionality. o Discuss any planned improvements.

Expected Operational Conditions & Testing

- **Environmental Conditions**: Indoor use, with standard room lighting and temperature.
- **Testing Methods**:
  - o Validate **RTC accuracy** by setting multiple reminders and checking trigger times. o Verify **LCD output** for displaying correct medicine information. o Test **buzzer and LED alerts** to ensure they activate at the correct times.
  - o Simulate a power failure to confirm system reliability (battery backup, RTC time retention). o Conduct user trials to assess ease of interaction (pressing buttons, reading alerts).

Key Talking Points & Anticipated Questions

- **How does the system ensure medication adherence?** o Answer: Uses **audio-visual cues** and a user-friendly interface for reminders.
- **Can it handle multiple medications for different times?** o Answer: Yes, the system can be programmed for multiple schedules.
- **What happens if the user doesn't acknowledge the reminder?**
  - o Answer: Currently, the alert continues until manually stopped. Future versions may notify caregivers remotely.
- **What are the future improvements?**
  - o Answer: IoT integration, automated pill dispensing, and a mobile app for remote monitoring.
- **How does the system handle power failures?**
  - o Answer: The **RTC module retains time** even during power loss, and battery backup can be integrated.

Video Demonstration

A short video showcasing:

- System setup and initial startup.
- Reminder scheduling and activation.
- User interaction and alert acknowledgment.
- Any optional IoT integration if implemented.

https://youtu.be/3SQGt3EJgs?si=mRyBTLWbfVurS0CJ

8. Project Deployment Considerations

Power Consumption Analysis

- **Energy Efficiency Measures**:
    - o Use **low-power components** (e.g., DS3231 RTC module). o Implement **sleep modes** on the Arduino to reduce power draw when idle.
    - o Consider a **rechargeable battery pack** to maintain operation during power failures.

Scalability & Future Enhancements

- **Enhancements for Larger-Scale Use**:
    - o Integration with **mobile apps** for remote monitoring. o Adding **voice alerts** for better accessibility (e.g., elderly users). o **Wi-Fi/IoT connectivity** for caregivers to track medication adherence remotely.
    - o Expanding to **automated pill dispensers** using a servo-controlled mechanism.

Real-World Deployment Challenges & Mitigation

- **Challenge: User Adoption & Accessibility** o **Mitigation**: Ensure a **simple interface**, large text on the LCD, and easy-to-press buttons.
- **Challenge: Reliability of Alerts** o **Mitigation**: Use **redundant alert mechanisms** (LED, buzzer, mobile notifications).
- **Challenge: Battery Life for Portable Use** o **Mitigation**: Optimize **power consumption** and explore **solar or rechargeable battery options**.
- **Challenge: Cost Constraints** o **Mitigation**: Use **affordable yet reliable components** and offer an option for **customization** based on user needs.

This ensures the project is **efficient, scalable, and practical for real-world application**.

9. Conclusion & Lessons Learned

Key Takeaways

I. **Simplicity is Key**: The project demonstrated that a minimalistic interface with clear feedback mechanisms is highly effective for medical reminder systems.

II. **Reliability**: The use of Arduino UNO ensured a robust and reliable system that can operate continuously without significant issues.

III. **User-Centric Design**: Incorporating snooze and acknowledgment buttons significantly improved user experience and adherence to medication schedules.

IV. **Challenges Faced**

V. **Power Management**: Ensuring the system operates efficiently on limited power was a challenge, especially for long-term use.

VI. **User Error Handling**: Designing intuitive error messages and recovery mechanisms required iterative testing and user feedback.

VII. **Hardware Limitations**: The limited memory and processing power of the Arduino UNO constrained the complexity of the system.

Knowledge Gained

I. **Hardware Integration**: Gained hands-on experience in integrating sensors, displays, and actuators with a microcontroller.

II. **User Interaction Design**: Learned the importance of designing for user convenience and accessibility, especially in medical applications.

III. **Problem-Solving**: Developed skills in troubleshooting hardware and software issues in embedded systems.

Recommendations for Future Projects

I. **Advanced Features**: Incorporate Wi-Fi or Bluetooth connectivity to enable remote monitoring and notifications via a mobile app.

II. **Enhanced Power Solutions**: Use rechargeable batteries or solar power to improve the system's sustainability.

III. **Scalability**: Consider using more powerful microcontrollers or IoT platforms for more complex applications, such as multi-user support or integration with electronic health records.

IV. **User Testing**: Conduct extensive user testing with diverse demographics to refine the interface and ensure usability for all potential users.

APPENDICES

Appendix A: Bill of Materials (BoM)

Appendix B: Firmware Libraries Used

Wire.h – For I2C communication with the RTC module

LiquidCrystal.h – For controlling the LCD display

https://drive.google.com/file/d/1Wm4BUNhhsLXU3b6sDTph70fJe57QPm51/view?usp=drivesdk https://github.com/Joveti/Medicine_Reminder_Project..git

REFERENCES;

Banzi, M., & Shiloh, M. (2022). *Getting started with Arduino* (4th ed.). Maker Media.

Brown, A. (2019). *Embedded systems design using microcontrollers*. Springer.

Osterberg, L., & Blaschke, T. (2005). Adherence to medication. *New England Journal of Medicine*, *353*(5), 487-497. https://doi.org/10.1056/NEJMra050100

Islam, S. M. R., Kwak, D., Kabir, M. H., Hossain, M., & Kwak, K. (2015). The internet of things for health care: A comprehensive survey. *IEEE Access*, *3*, 678-708. https://doi.org/10.1109/ACCESS.2015.2437951